Greek fortition in Logical Phonology [*]

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Logical Phonology (henceforth, LP; e.g., Bale et al. 2014, Bale and Reiss 2018, Reiss 2021, Dabbous et al. 2024, Gorman and Reiss 2024, Reiss 2024) is a formally rigorous, *substance-free* theory of phonology. LP posits that features are binary-valued (rather than privative) and segments are thus sets of valued features linked to a X-slot. A segment must be *consistent*: if a segment ς is specified +F, it cannot also be specified −F or vice versa. However, a segment can be *underspecified*: ς need not contain any specification for F at all. Following Poser (1982) et seq., the distinction between feature-filling and feature-changing rules is derived by decomposing feature-changing processes (e.g., voice assimilation) into separate, independently motivated feature insertion (*unification*) and deletion (*subtraction*) operations.

(1) Unification: If A and B are feature specifications, then $A \sqcup B = A \cup \{c\text{F} \mid c\text{F} \in B \land -c\text{F} \notin A\}$.

(2) Subtraction: If A and B are feature specifications, then $A \setminus B = \{c\text{F} \mid c\text{F} \in A \land c\text{F} \notin B\}$.

Unification provides a novel solution to the decades-old problem (e.g., Lees 1961:12–14, Lightner 1971:236, Archangeli 1988, Bonet 1995) of targeting relatively underspecified structures without also affecting more richly specified ones. It is also a principled way to implement Sharon Inkelas and colleagues' notion of *inalterability as prespecification* (e.g., Inkelas and Cho 1993, Kiparsky 1993, Buckley 1994, Inkelas 1995, Inkelas and Orgun 1995, Inkelas et al. 1997), according to which

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segments underspecified for some feature F are *mutable* with respect to rules adding a ±F, allowing many patterns—previously analyzed using morphophonological rules, morpheme-specific constraints, cophonologies, or lexical diacritics—to be generated by the narrow phonology.

Markopoulos and Revithiadou (2023), henceforth M&R, describe a process of manner dissimilation in Modern Greek by which certain labial fricatives, those usually spelled ⟨φ, β⟩ as in (3ab), strengthen—and devoice in the case of ⟨β⟩—to [p] before the sibilant /s/; they are realized as [f, v] elsewhere. However, other labial fricatives, usually spelled ⟨υ⟩, resist fortition in this context.[1]

(3) Fortition (M&R; additional data from Despina Nifakos, p.c.):

	IPFV.1SG /-o/	PST-1SG /-s-a/	
a.	[ˈɣra**f**o]	[ˈɣra**p**sa]	‘write’
	[ˈva**f**o]	[ˈeva**p**sa]	‘paint’
	[ˈste**f**o]	[ˈeste**p**sa]	‘crown’
b.	[ˈɾa**v**o]	[ˈɾa**p**sa]	‘sew’
	[ðuˈle**v**o]	[ˈðule**p**sa]	‘work’
	[peˈðe**v**o]	[ˈpeðe**p**sa]	‘annoy’
c.	[vraˈve**v**o]	[vraˈve**f**sa]	‘award’
	[ermiˈne**v**o]	[erˈmine**f**sa]	‘interpret’
	[peˈðe**v**o]	[peˈðe**f**sa]	‘instruct’

Note especially the homonym ⟨παιδεύω⟩ [peˈðevo], which means ‘annoy’ if it undergoes fortition and stress shift in the past tense, and ‘instruct’ if it does not.

M&R cite a number of acoustic studies confirming phonetic equivalences and conduct a formal acceptability judgment task ($n = 35$); they then analyze this pattern using Gradient Harmonic Grammar (Smolensky and Goldrick 2016). However, this data can also be generated using the more modest architectural assumptions of LP. Suppose that *inalterable* /v/ is underlyingly +CONTINUANT /v/, whereas *mutable* /F, V/ denote labial fricatives underspecified for this feature.

[1]Abbreviations used: IPFV: imperfective; PST: past; 1SG: first-person singular. Some past-tense forms also show an prothetic [e] and/or stress-shift in the past tense.

(4) Feature specification (partial):

	/F/	/V/	/v/
CONTINUANT			+
VOICE	−	+	+

(5), a unification rule, strengthens /F, V/ when followed by a sibilant, but applies vacuously to inalterable /v/, which is already specified for CONTINUANT. It is similarly vacuous when applying to other labial consonants, such as /p, b, m/. A later context-free redundancy rule specifies /F, V/ not targeted by (5) as fricatives, and similarly applies vacuously to fully-specified segments including other labials. For completeness, the voice assimilation rules (implemented as as a subtraction followed by a unification), critical orderings, and sample derivations are also provided.

(5) Fortition: $\left[+\textsc{Labial}\right] \sqcup \left\{-\textsc{Continuant}\right\} \;/\; \_\_ \left[+\textsc{Strident}\right]$

(6) Default frication: $\left[+\textsc{Labial}\right] \sqcup \left\{+\textsc{Continuant}\right\}$

(7) Voice assimilation (part 1): $\left[-\textsc{Sonorant}\right] \setminus \left\{+\textsc{Voice}\right\} \;/\; \_\_ \begin{bmatrix}-\textsc{Voice}\\ -\textsc{Sonorant}\end{bmatrix}$

(8) Voice assimilation (part 2): $\left[-\textsc{Sonorant}\right] \sqcup \left\{-\textsc{Voice}\right\} \;/\; \_\_ \begin{bmatrix}-\textsc{Voice}\\ -\textsc{Sonorant}\end{bmatrix}$

(9) Critical orderings: (5) $\ll$ (6); (7) $\ll$ (8)

(10) Sample derivations (stress omitted):

UR:	ɣraF-o	ɣraF-s-a	ɾaV-o	ɾaV-s-a	vravev-o	vravev-s-a
Rule (5):		ɣra**p**sa		ɾa**b**so		
Rule (6):	ɣra**f**o		ɾa**v**o			
Rules (7–8):				ɾa**p**sa		vrave**f**sa
SR:	ɣrafo	ɣrapsa	ɾavo	ɾapsa	vravevo	vravefsa

LP provides all necessary representations and operations needed to implement Greek's "exceptional" fortition pattern, previously analyzed by M&R using the much richer architectural primitives of Gradient Harmonic Grammar.

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